

SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)
DepartmentofComputerScienceandEngineering

ASSESSMENTTOOL3–MCQTest

CourseCode:	CSA0305	CourseTitle:	DataStructures
COAssessed:	CO3–Precision(BL3,BL5)	Assessment:	AssessmentTool3–MCQTest
Weightage:	30%ofCIA	TotalMarks:	100
CO3Statement:	<i>SolveproblemsforoptimizeddatastorageandretrievalusingBinarySearchTrees,AVLTrees,B-Trees, Red-Black Trees, and Splay Trees.</i>		
Student Name:	M. VAISHNAV	Reg. No.:	192524251
Section/Batch:		Date:	12/08/2026

CodeofConduct

I M. VAISHNAV (192524251) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M. VAISHNAV (192524251)

Instructions

- Thistestcontains25MCQTestquestions(4markseach),Total:100 Marks.
- When a student answer using MCQ, in evaluation the answer should be with following requirements:
Identifythemostappropriateanswer,conceptualunderstanding,problem-solvingability,applicationof knowledge.
- Nonenegativemarking.Unansweredquestionscarryzeromarks.

Section/Topic	Focus	QNos	Marks
Preliminaries–TreeTerminology	Root,height,level,siblingsandsubtree	1–2	8
BinaryTree, BinarySearchTree	Properties,binarytreeandbinarysearchtree,in sersion and deletion	3–9	28
Tree Traversals	Traversaltypes,Inorder,PostorderandPreorder	10–14	20
AVL Tree	Balancingfactor,Rotation,Typesofrotation,examples	15–16	8
ApplicationsofSearchTrees	Usageofbinarytree	17	4
BTree -2-3tree,2-3-4tree	Properties,ExampleandOperations	18–19	8
TRIE	Properties,structuresandOperations	20	4
RedBlackTree	Properties,BalancingandOperations	21–22	8
SplayTree	Splaying,Typesandexamplesforinsertionanddeletion	23–25	12
GRANDTOTAL:			100Marks

MCQ

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	20	Demonstrate thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Show good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Application Skills	20	Correctly applies concepts to solve complex and scenario-based MCQs	Applies concepts correctly in most moderate-level questions	Struggles with application in complex scenarios	Unable to apply concepts effectively
Analytical Thinking	30	Consistently selects correct answers for high-order questions	Performs well in moderate analytical questions	Limited success in analytical questions	Unable to interpret analytical/problem-based questions
Time Management	20	Completes test efficiently with time for review	Completes test within time	Struggles to complete test on time	Unable to complete test
Accuracy of Responses	10	90–100% correct answers	75–89% correct answers	50–74% correct answers	Below 50% correct answers

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



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QUESTIONS

- Q1**
Tree Traversal
4 marks

✓
- Q2**
Binary Search Tree
4 marks

✓
- Q3**
Red-Black Tree
4 marks

✓
- Q4**
Red-Black Tree
4 marks

✓
- Q5**
B-Tree
4 marks

✓
- Q6**
Tree Terminology
4 marks

✓
- Q7**
Tree Terminology
4 marks

✓
- Q8**
Binary Search Tree
4 marks

✓
- Q9**
Binary Tree
4 marks

✓
- Q10**
Binary Tree
4 marks

✓
- Q11**
Binary Tree
4 marks

✓
- Q12**
Binary Search Tree
4 marks

✓
- Q13**
Tree Traversals
4 marks

✓
- Q14**
Tree Traversals
4 marks

✓
- Q15**
Tree Traversals
4 marks

✓
- Q16**
Tree Traversals
4 marks

✓
- Q17**
AVL Tree
4 marks

✓
- Q18**
AVL Tree
4 marks

✓
- Q19**
Search Tree
4 marks

✓
- Q20**

✓

Q1 Tree Traversal

4 marks

You analyze the post-order traversal of a tree representing project tasks: [D, E, B, F, G, C, A].
You want to know

A. 3

B. 4

C. 5

D. 2

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QUESTIONS

- Q1**
Tree Traversal
4 marks
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4 marks
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B-Tree
4 marks
- Q6**
Tree Terminology
4 marks
- Q7**
Tree Terminology
4 marks
- Q8**
Binary Search Tree
4 marks
- Q9**
Binary Tree
4 marks
- Q10**
Binary Tree
4 marks
- Q11**
Binary Tree
4 marks
- Q12**
Binary Search Tree
4 marks
- Q13**
Tree Traversals
4 marks
- Q14**
Tree Traversals
4 marks
- Q15**
Tree Traversals
4 marks
- Q16**
Tree Traversals
4 marks
- Q17**
AVL Tree
4 marks
- Q18**
AVL Tree
4 marks
- Q19**
Search Tree
4 marks
- Q20**

Q2 Binary Search Tree

4 marks

In a Binary Search Tree, the left child contains:

- A. Greater values
- B. Smaller values
- C. Equal values
- D. Random values

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QUESTIONS

Q1
Tree Traversal
4 marks

Q2
Binary Search Tree
4 marks

Q3
Red-Black Tree
4 marks

Q4
Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q7
Tree Terminology
4 marks

Q8
Binary Search Tree
4 marks

Q9
Binary Tree
4 marks

Q10
Binary Tree
4 marks

Q11
Binary Tree
4 marks

Q12
Binary Search Tree
4 marks

Q13
Tree Traversals
4 marks

Q14
Tree Traversals
4 marks

Q15
Tree Traversals
4 marks

Q16
Tree Traversals
4 marks

Q17
AVL Tree
4 marks

Q18
AVL Tree
4 marks

Q19
Search Tree
4 marks

Q20

Q3 Red-Black Tree

4 marks

During insertion, when a red node has a red parent and the uncle is also red, what action is taken:

A. Single rotation

B. Double rotation

C. Recolor parent and uncle black, grandparent red

D. No action needed

Save Answer

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QUESTIONS

Q1
Tree Traversal
4 marks

Q2
Binary Search Tree
4 marks

Q3
Red-Black Tree
4 marks

Q4
Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q7
Tree Terminology
4 marks

Q8
Binary Search Tree
4 marks

Q9
Binary Tree
4 marks

Q10
Binary Tree
4 marks

Q11
Binary Tree
4 marks

Q12
Binary Search Tree
4 marks

Q13
Tree Traversals
4 marks

Q14
Tree Traversals
4 marks

Q15
Tree Traversals
4 marks

Q16
Tree Traversals
4 marks

Q17
AVL Tree
4 marks

Q18
AVL Tree
4 marks

Q19
Search Tree
4 marks

Q20

Q4 Red-Black Tree

4 marks

Which property ensures that no two red nodes are adjacent in a Red-Black Tree?

A. Root is black

B. Red nodes have black children

C. Every path from root to leaf has the same number of black nodes

D. Leaves are red

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QUESTIONS

Q1 Tree Traversal
4 marks

Q2 Binary Search Tree
4 marks

Q3 Red-Black Tree
4 marks

Q4 Red-Black Tree
4 marks

Q5 B-Tree
4 marks

Q6 Tree Terminology
4 marks

Q7 Tree Terminology
4 marks

Q8 Binary Search Tree
4 marks

Q9 Binary Tree
4 marks

Q10 Binary Tree
4 marks

Q11 Binary Tree
4 marks

Q12 Binary Search Tree
4 marks

Q13 Tree Traversals
4 marks

Q14 Tree Traversals
4 marks

Q15 Tree Traversals
4 marks

Q16 Tree Traversals
4 marks

Q17 AVL Tree
4 marks

Q18 AVL Tree
4 marks

Q19 Search Tree
4 marks

Q20

Q5 B-Tree

4 marks

A 2-3-4 tree contains a 4-node with keys [20, 40, 60]. If a new key 50 is to be inserted, what is the result of the split?

A. New root becomes [40]; children: [20], [50], [60]

B. Root stays [20, 40, 50, 60]

C. Keys rearranged into a single 4-node

D. Insertion is not possible

Save Answer

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QUESTIONS

Q1 Tree Traversal
4 marks

Q2 Binary Search Tree
4 marks

Q3 Red-Black Tree
4 marks

Q4 Red-Black Tree
4 marks

Q5 B-Tree
4 marks

Q6 Tree Terminology
4 marks

Q7 Tree Terminology
4 marks

Q8 Binary Search Tree
4 marks

Q9 Binary Tree
4 marks

Q10 Binary Tree
4 marks

Q11 Binary Tree
4 marks

Q12 Binary Search Tree
4 marks

Q13 Tree Traversals
4 marks

Q14 Tree Traversals
4 marks

Q15 Tree Traversals
4 marks

Q16 Tree Traversals
4 marks

Q17 AVL Tree
4 marks

Q18 AVL Tree
4 marks

Q19 Search Tree
4 marks

Q20

Q6 Tree Terminology

4 marks

A subtree of a tree is:

A. A separate graph

B. A part of the tree that is itself a tree

C. Only leaf nodes

D. Only internal nodes

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Red-Black Tree
4 marks

Q4
Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q7
Tree Terminology
4 marks

Q8
Binary Search Tree
4 marks

Q9
Binary Tree
4 marks

Q10
Binary Tree
4 marks

Q11
Binary Tree
4 marks

Q12
Binary Search Tree
4 marks

Q13
Tree Traversals
4 marks

Q14
Tree Traversals
4 marks

Q15
Tree Traversals
4 marks

Q16
Tree Traversals
4 marks

Q17
AVL Tree
4 marks

Q18
AVL Tree
4 marks

Q19
Search Tree
4 marks

Q20

Q8 Binary Search Tree

4 marks

How many distinct binary search trees can be created out of 4 distinct keys?

A. 4

B. 42

C. 24

D. 14

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Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q7
Tree Terminology
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Q8
Binary Search Tree
4 marks

Q9
Binary Tree
4 marks

Q10
Binary Tree
4 marks

Q11
Binary Tree
4 marks

Q12
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Q13
Tree Traversals
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Q14
Tree Traversals
4 marks

Q15
Tree Traversals
4 marks

Q16
Tree Traversals
4 marks

Q17
AVL Tree
4 marks

Q18
AVL Tree
4 marks

Q19
Search Tree
4 marks

Q20

Q10 Binary Tree

4 marks

A complete binary tree is one in which:

- A. All levels are fully filled
- B. Every node has two children
- C. All levels except possibly the last are completely filled
- D. Only root is filled

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4 marks

Q2
Binary Search Tree
4 marks

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Red-Black Tree
4 marks

Q4
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Q5
B-Tree
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Q16
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4 marks

Q17
AVL Tree
4 marks

Q18
AVL Tree
4 marks

Q19
Search Tree
4 marks

Q20
B-Tree
4 marks

Q21
Trie
4 marks

Q22

retrieval, you check the number of edges between node 10 and the root. How many edges are in the path from node 10 to the root?

A. 1

B. 3

C. 2

D. 5

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Tree Traversals
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Q17
AVL Tree
4 marks

Q18
AVL Tree
4 marks

Q19
Search Tree
4 marks

Q20
B-Tree
4 marks

Q21
Trie
4 marks

Q22

using traversal logs: In-order and post-order. Given: In-order: D B E A F C Post-order: D E B F C A What is the root

A. A

B. D

C. B

D. C

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Q14
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Q15
Tree Traversals
4 marks

Q16
Tree Traversals
4 marks

Q17
AVL Tree
4 marks

Q18
AVL Tree
4 marks

Q19
Search Tree
4 marks

Q20

Q24 Splay Tree

4 marks

Which rotation is used when the node is the direct child of the root?

20

/

10

A. Zig

B. Zig-Zig

C. Zag-Zig

D. Split

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